

6 (a) State what is meant by simple harmonic motion.

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..... [2]

(b) A length of thick copper wire is formed into a rectangular frame, as shown in Figure 6.1.

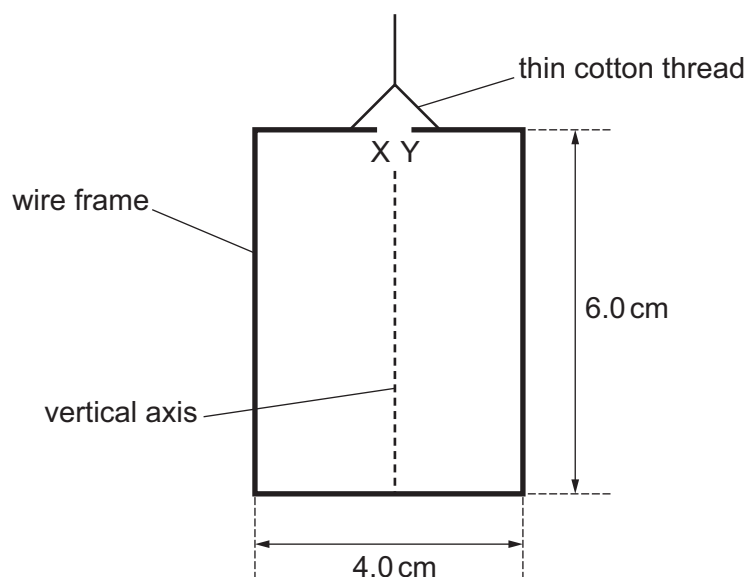


Figure 6.1

The frame has side lengths of 6.0 cm and 4.0 cm.
There is a small gap XY at the ends of the wire at the top of the frame.

The frame is suspended so that it hangs vertically using thin cotton thread attached to the two ends of the wire. The frame oscillates about its vertical axis, as shown in the view from above in Figure 6.2.

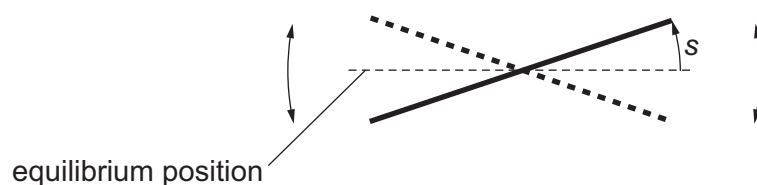


Figure 6.2

The oscillations are simple harmonic. The displacement s of the edge of the frame from its equilibrium position varies with time t according to

$$s = 3.2 \sin 6.9t$$

where s is in mm and t is in s.

- (i) Calculate the period T of the oscillations.

$$T = \dots\dots\dots \text{ s } [2]$$

- (ii) Show that the maximum speed of the edges of the frame is 22 mm s^{-1} .

[2]

- (c) The frame in **6(b)** is in a uniform horizontal magnetic field that is parallel to the equilibrium position of the frame.

The flux density of the magnetic field is 0.047 T .

- (i) Calculate the electromotive force (e.m.f.) E_0 induced across XY when the frame is moving at its maximum speed.

$$E_0 = \dots\dots\dots \text{ V } [3]$$

- (ii) On Figure 6.3, sketch the variation of the induced e.m.f. E across XY with t between $t = 0$ and $t = 2T$.

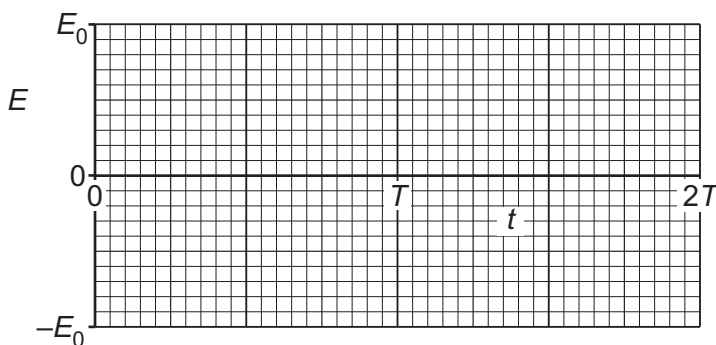


Figure 6.3

[2]